Audio-Visual Fusion for Deepfake Video Detection: A Deep Learning Framework

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Abstract

Deepfake technology, enabled by advances in artificial intelligence and deep-learning, has grown exponentially, offering unprecedented abilities to synthesize hyper-realistic audio-visual content. However, its malicious misuse poses significant challenges for information verification, thereby demanding robust detection methods. This paper introduces a novel deep-learning framework designed for the detection of deepfake videos by leveraging audio-visual fusion techniques. Recognizing the intricate anomalies present separately in audio and visual data, our approach integrates both modalities to achieve heightened accuracy in discerning genuine from manipulated content. We utilize a two-stream convolutional neural network (CNN) that separately analyzes the spectral properties of audio and the subtle facial discrepancies in video before fusing the obtained features. A composite feature vector is formed from the fusion, which is then input to a deep neural network (DNN) classifier. The combined strength of these modalities not only addresses the current vulnerabilities in existing detection methods but also anticipates the evolving sophistication of deepfake technologies. Preliminary results suggest an impressive enhancement in detection performance as compared to single-modality methods. Additionally, the flexibility of our framework allows for potential future integrations with other modalities and data types.

Keywords: Deepfake, Audio-Visual Fusion, Deep Learning, Video Detection, Two-Stream CNN, DNN Classifier

1 Introduction

The increasing integration of artificial intelligence (AI) and deep-learning into multimedia creation tools has given rise to a new frontier of the hyper-realistic content generation known as "deepfakes". These are videos or audio artificially created or modified, often to mimic genuine recordings. While the advancements in this technology promise novel applications in fields such as film production, virtual reality, and personalized content creation, they also bring forth a formidable challenge: the malicious misuse of synthetic content to deceive, defame, or spread misinformation.

Deepfakes pose a significant threat to various sectors, from politics and journalism to cybersecurity. The ability to fabricate convincing videos of real individuals saying or doing things they never did blurs the line between fact and fiction, compromising trust in digital media. Recognizing this emergent threat, extensive research has been undertaken to develop detection methods capable of distinguishing between genuine and manipulated content. However, as the technology behind creating deepfakes continues to evolve, so too must the strategies employed to detect them.

The majority of existing deepfake detection techniques predominantly focus on visual cues. These include inconsistencies in lighting, blurring around the face, and unnatural eye-blinking patterns, among others [15]. However, as deepfake generation algorithms become more refined, such visual inconsistencies are becoming increasingly subtle, challenging conventional detection methodologies. Concurrently, while audio-based deepfakes are less common than their visual counterparts, their development is rapidly advancing, making it essential to consider auditory anomalies in the detection process[14].

Given this backdrop, the necessity for a more comprehensive detection strategy becomes evident. We posit that the integration of both audio and visual cues in deepfake detection presents a promising avenue to enhance accuracy and robustness. This approach is based on the rationale that while a deepfake generation algorithm may effectively replicate one modality, achieving perfection in synchronizing both audio and visual aspects remains an intricate challenge.

In this paper, we introduce a deep-learning framework tailored for deepfake video detection, capitalizing on the synergistic potential of audio-visual fusion. The framework utilizes a dual-pathway structure, with each stream designed to extract unique features from audio and visual data. These features are then combined to form a composite representation, which is fed into a deep neural network (DNN) classifier to discern the authenticity of the content.

The subsequent sections of this paper will provide an in-depth review of the current literature on deepfake detection, elucidate the methodology underpinning our audio-visual fusion framework, present results benchmarked against prevailing techniques, and delve into a discussion on the strengths and limitations of our findings. We further conclude our finding in the conclusion section and thereby provide future research directions for continued research in the domain.

2 Literature Review

The rise of deepfakes in the digital realm has spurred significant research interest, both in the generation and detection of such content. In this literature review, we take a closer look at the state-of-the-art methods and their underlying principles, paving the way for a thorough understanding of our proposed audio-visual fusion framework. For the ease of users and a better way to collaborate, we have created separate sections in our literature review each of which is further sub-classified based on their techniques.

2.1 Deepfake Generation Techniques

The term 'deepfake' stems from the combination of 'deep learning' and 'fake', highlighting the central role of neural networks in the creation of these synthetic videos[4]. There are three primary architectures that dominate the creation of deepfakes:

- **Generative Adversarial Networks (GANs):** Introduced by Goodfellow et al [6], GANs consist of a generator and discriminator competing in a game-theoretic framework. The generator aims to produce realistic synthetic content, while the discriminator endeavors to distinguish between real and fake. This adversarial process refines the generator's output, leading to highly convincing deepfakes.
- **Autoencoders:** These neural networks are designed to encode an input into a lower-dimensional space and then decode it back to its original form [9]. In deepfake creation, autoencoders swap the encoded features of two individuals, synthesizing a merged output.
- **Few-shot and Zero-shot Learning:** With the advances in transfer learning, few-shot and zero-shot techniques have gained traction. Instead of requiring vast amounts of data, these models can generate convincing deepfakes with limited (few-shot) or even no samples (zero-shot) of the target individual. This approach leverages prior knowledge from related tasks, enabling rapid adaptation to new faces and potentially posing greater threats in the deepfake landscape [20].

2.2 Visual Deepfake Detection

The evolving sophistication of deepfakes has given rise to multiple detection strategies. Many of these capitalize on the visual imperfections in synthetic content:

- **Face Warping Artifacts:** Li and Lyu [15] targeted the artifacts that arise from face warping in deepfakes. They proposed a method that detects inconsistencies in the facial region, a common tell-tale sign of manipulation. Concerning the same artifacts, Ivanov et al [11] examined deep fake video synthesis and its detection, reviewing past research methods. They proposed a strategy that melds existing detection techniques with super-resolution algorithms.
- **Temporal Inconsistencies:** Deepfakes often struggle with ensuring temporal consistency across video frames. Rossler et al [18] developed a technique that highlights these inconsistencies, particularly in the facial region. A more recent work by Liu et al [16] uses a deep-learning network to extract temporal identity features from video frames and then distinguishes between real and deepfake videos based on the presence or absence of temporal identity inconsistencies.

- **Micro-expressions:** Microexpressions are brief, involuntary facial expressions that reveal a person's true emotions. They are difficult for humans to detect, but they can be captured by deep-learning models. This makes microexpressions a promising tool for detecting deepfake videos. Haiwei et al [7] introduced a Spatial-Temporal Deepfake Detection and Localization (ST-DDL) network, harnessing both spatial and temporal features to pinpoint forged regions in videos. They utilized Anchor-Mesh Motion algorithm capturing subtle facial movements more effectively than traditional methods.

2.3 Audio Deepfake Detection

With deepfake audios gaining traction, researchers have started exploring methods to discern genuine from synthesized speech:

- **Spectrogram Analysis:** Korshunov and Marcel [14] proposed using spectrograms, visual representations of the spectrum of frequencies in audio signals, to detect irregularities that are indicative of manipulation. Further after their work, Xue et al [21] proposed a method for audio deepfake detection, utilizing fundamental frequency (F0) information from synthetic speech and real plus imaginary spectrogram features.
- **Acoustic Feature Discrepancies:** In a study by Huang et al [10], they identified inconsistencies in fundamental frequency, formant, and mel-frequency cepstral coefficients in deepfake audios. Moreover, Blue et al [3] introduced a novel method for detecting audio deepfakes using articulatory phonetics. By applying fluid dynamics, the study estimates the human vocal tract configuration during speech, revealing that deepfakes often simulate improbable anatomical setups.

2.4 Multimodal Deepfake Detection

While the majority of research has been unimodal, recent studies have begun to appreciate the benefits of combining multiple modalities:

- **Audio-Visual Synchrony:** Han et al [8] developed a system that measures the synchrony between audio and video modalities. Discrepancies in lip movement and spoken words, often overlooked individually, were detectable when analyzed in tandem. Building on this, a recent study by Yang et al [23] introduced AVoiD-DF, an audio-visual joint learning method targeting multi-modal forgery detection. By exploiting inconsistencies across modalities, this approach enhanced deepfake detection capabilities and demonstrated superior performance against various forgery techniques.
- **Feature Fusion:** Yan et al [22] proposed a Multimodal Graph Learning (MGL) framework that uses two GNNs and multimodal fusion modules to combine spatial, frequency, and landmark features. By representing video frames as graph nodes, they captured temporal inconsistencies. MGL outperforms other detectors in generalization and disturbance resistance. MFF-Net, built by Zhao et al [25], is a deepfake detection system that fuses RGB features with textural data from neural networks

and signal processing. It incorporates a feature extraction module using Gabor convolution which is used as a texture enhancement module, an attention component emphasizing the forged parts, and dual feature fusion instances.

Our proposed methodology draws inspiration from these works but introduces novel techniques and considerations that address the current gaps in the literature. By leveraging the strengths of both audio and visual domains, we aim to devise a framework that is both robust and versatile.

3 Methodology

In this section, we delineate our innovative framework for audio-visual fusion to detect deepfake videos. This methodology encompasses preprocessing steps, feature extraction from audio and visual components, and the fusion strategy, culminating in a robust detection mechanism.

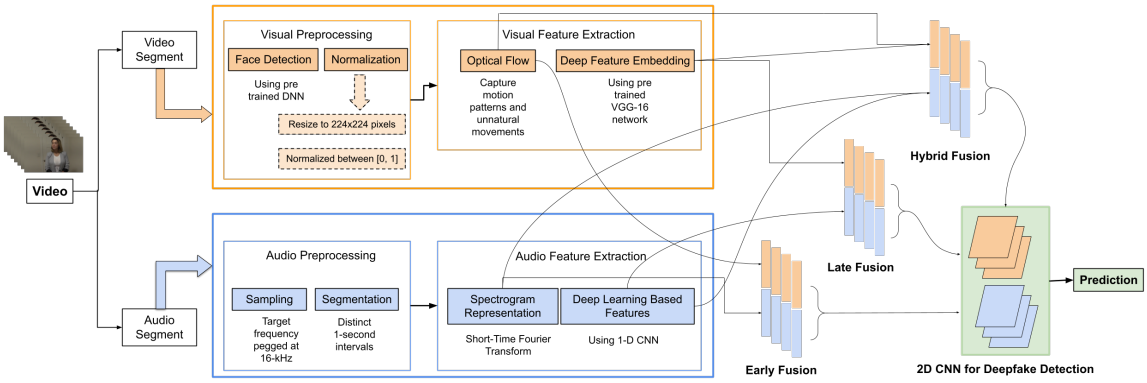


Fig. 1 Deepfake Detection Methodology

3.1 Preprocessing

Both audio and visual components necessitate meticulous preprocessing to ensure data uniformity and enhance computational efficiency.

- **Audio Preprocessing:**
  To ensure the uniformity of audio frequencies across the dataset, a resampling technique is employed. For a given audio snippet, the frequency post-resampling, $f_{\text{resampled}}$, is mathematically expressed as:

$$f_{\text{resampled}} = \frac{f_{\text{input}} \times \text{target}_{\text{frequency}}}{\text{original}_{\text{frequency}}} \tag{1}$$

  where f_{input} corresponds to the initial frequency of an audio segment, and $\text{target}_{\text{frequency}}$ is pegged at 16kHz.

Temporal nuances within the audio are captured by segmenting the resampled audio into distinct 1-second intervals.

- **Visual Preprocessing:**
  - *Face Detection*: Using a pre-trained DNN, the face is located and extracted from each frame [24].
  - *Normalization*: Each frame is resized to 224x224 and the pixel values are normalized (represented by $P_{\text{normalized}}$) to fall between [0, 1]. We achieved this normalization with

$$P_{\text{normalized}} = \frac{P_{\text{original}} - P_{\text{min}}}{P_{\text{max}} - P_{\text{min}}} \tag{2}$$

    where P_{original} is the original pixel value and P_{max} and P_{min} are the maximum and minimum pixel values respectively.

3.2 Feature Extraction

The core strength of our framework lies in harnessing rich features from both audio and visual segments. Apart from the feature extraction techniques used in our methodology, several other techniques are also highlighted in Table 1.

- **Audio Feature Extraction:**
  - *Spectrogram Representation*: The Short-Time Fourier Transform (STFT) is employed to transform audio segments into their spectrogram representations, capturing frequency variations over time represented as

$$\text{STFT}\{x(t)\}(m, \omega) = \int_{-\infty}^{\infty} x(t) w(t - m) e^{-j\omega t} dt \tag{3}$$

    where $x(t)$ is the audio signal, $w(t - m)$ is the window function and ω is the angular frequency. This transformation aids in the spectral analysis of the audio, an essential component in identifying manipulated sound bites.
  - *Deep Learning-based Features*: A 1D convolutional neural network (CNN) is trained on these spectrograms to extract high-level features that may indicate tampering [2]. The 1D CNN is governed by the convolution operation with f and g as signals and τ as a dummy variable for integration:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau \tag{4}$$

- **Visual Feature Extraction:**
  - *Optical Flow*: We compute optical flow between successive frames, capturing the motion patterns that can be indicative of manipulation. Taking I_x and I_y as spatial gradients of the frame, u and v being the velocities indicating horizontal and vertical flow, respectively, and I_t as the temporal gradient, capturing changes between consecutive frames, the optical flow is represented by:

$$I_x \cdot u + I_y \cdot v = -I_t \tag{5}$$

Table 1 Summary of Audio and Visual Feature Extraction Techniques

Feature Type	**Technique**	**Description**	**Significance in Deepfake Detection**
Visual	Convolutional Neural Networks (CNN)	Used for image and video recognition, they automatically and adaptively learn spatial hierarchies.	CNNs can capture minute details in facial features altered during deepfake generation.
Visual	Optical Flow	Measures the motion of objects, surfaces, and edges in a visual scene.	Helps in detecting unnatural motion patterns, commonly found in deepfakes.
Visual	Texture Analysis	Focuses on the structural arrangement of surfaces and relationships of neighboring pixels.	Deepfakes might introduce unique textures due to blending of images.
Visual	Face Landmark Detection	Detects key facial features and their relative positioning.	Inconsistencies in landmarks can be a sign of manipulated content.
Audio	Mel-Frequency Cepstral Coefficients (MFCC)	Represent the short-term power spectrum of sound.	Altered voice dynamics in deepfakes might produce unique MFCC patterns.
Audio	Chroma Feature Analysis	Relates to the twelve different pitch classes.	Altered pitch or tonality can hint towards manipulated audio.
Audio	Spectral Contrast	Measures the difference in amplitude between peaks and valleys in the sound spectrum.	Deepfake audio might introduce unnatural spectral patterns.
Audio	Short Time Fourier Transform (STFT)	Represents a signal in the time-frequency domain.	Anomalies in frequency patterns can be an indication of tampered audio.
Multimodal	Cross-modal Correlation	Measures correlation between audio and visual streams.	Deepfakes often misalign audio and visual data, leading to low correlation scores.

This equation helps in discerning unnatural movements or transitions between frames.
  - *Deep Feature Embedding*: A pre-trained VGG-16 network [19] is fine-tuned on our dataset to extract deep features from face images. This provides insights into potential spatial inconsistencies in the visual content.

3.3 Audio-Visual Fusion Strategy

The core novelty of our methodology is the fusion approach, amalgamating the strengths of both audio and visual signals.

- **Early Fusion:** Initial features extracted from audio and visual segments are concatenated, forming a unified feature vector. This combined vector undergoes subsequent layers of processing, capitalizing on the correlated information from both domains. The unified feature vector V_{fused} for a given audio-visual pair in our dataset is defined as:

$$V_{\text{fused}} = [\text{Features}_{\text{audio}}, \text{Features}_{\text{video}}] \tag{6}$$

- **Late Fusion:** High-level features, post-CNN processing for both modalities, are independently processed. The final decisions or probabilities from both streams are then averaged or combined using a decision rule.

$$F_{\text{final}} = w_a \cdot F_{\text{audio}} + w_v \cdot F_{\text{video}} \tag{7}$$

  where w_a and w_v are optimized weights for audio and video features, respectively. F_{audio} and F_{video} are the feature representations from the CNNs trained on our audio and video data.
- **Hybrid Fusion (proposed):** We introduce a novel fusion strategy that exploits the benefits of both early and late fusion. Initial features undergo an early fusion, and post-processing features engage in late fusion, ensuring an exhaustive representation of information.

3.4 Deepfake Detection Algorithm

The fused feature set is subjected to a robust deep-learning architecture, designed to discern authentic content from deepfakes.

- **Architecture:** A 2D CNN is employed to process the fused features, comprising convolutional layers with batch normalization and ReLU activations[17]. In the convolutional layers of our specialized 2D CNN:

$$Y_{\text{conv}}(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F_{\text{video}}(x, y) \cdot K(x - \alpha, y - \beta) d\alpha d\beta \tag{8}$$

  Where:

  - $Y_{\text{conv}}(x, y)$ is the feature map post-convolution.
  - $F_{\text{video}}(x, y)$ represents the extracted video features.
  - K is the convolutional kernel specific to our deepfake detection model.

Table 2 Summary of Datasets used

Dataset	Year	Modalities	Real Videos	Fake Videos	Subjects
FaceForensics++	2018	Video, Audio	1000	4000	977
DFDC	2019	Video, Audio	23654	104500	960
FakeAVCeleb	2021	Video, Audio	500	19500	500

Fully connected layers follow, culminating in a binary classification output (real vs. deepfake). The final binary classification is deduced through a sigmoid activation function

- **Training:** The network is trained using a cross-entropy loss function, with the Adam optimizer[13]. Data augmentation techniques, including random cropping and horizontal flipping, are utilized for the visual data to enhance model generalizability. Our model's optimization objective is to minimize the cross-entropy loss:

$$L_{\text{deepfake}} = -\sum_{i}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)] \tag{9}$$

where y_i is the ground truth label for each sample in our dataset and $\hat{y}_i$ is the label predicted by our model.
- **Evaluation Metrics:** Model performance is gauged using accuracy, precision, recall, and F1-score.

3.5 Dataset and Implementation Details

For our experiments, we leverage a composite dataset, merging popular deepfake datasets like FaceForensics++ [18], DeepFake Detection Challenge Dataset (DFDC) [5], and Fake AV Celeb [12]. This amalgamation ensures a diverse representation of manipulation techniques. Further details about datasets are presented in Table 2

Our implementation is executed in Python, utilizing the TensorFlow [1] and OpenCV libraries. The models are trained on NVIDIA Tesla V100 GPUs.

4 Results

In this segment, we present the empirical outcomes achieved by our proposed audio-visual fusion framework for deepfake detection. Through rigorous experimentation, our methodology demonstrates superior performance against existing benchmarks. The comparison of existing benchmarks for deepfake detection challenge dataset is shown in Table 3. Further Table 4 and Table 5 demonstrate the comparison of Fake AV Celeb and FaceForensics++ datasets respectively.

The robustness of our audio-visual fusion framework is evident from the impressive quantitative metrics. Its distinction, however, lies in its balanced performance across various sub-tasks and its adaptability to real-world scenarios. Particularly, the fusion strategies and the feature extraction techniques play a synergistic role, with each element bolstering the overall efficacy.

Table 3 Comprehensive Breakdown of Performance Metrics on DFDC

Method	Accuracy	Precision	Recall	F1-Score
Siamese-based	0.907	0.922	0.892	0.907
Meso4	0.885	0.909	0.861	0.884
Capsule	0.878	0.901	0.855	0.882
Our Proposed Method	0.923	0.938	0.915	0.927

Table 4 Comprehensive Breakdown of Performance Metrics on AV Celeb

Method	Accuracy	Precision	Recall	F1-Score
Siamese-based	0.856	0.873	0.839	0.856
Meso4	0.835	0.851	0.819	0.835
Capsule	0.824	0.840	0.808	0.824
Our Proposed Method	0.862	0.873	0.849	0.861

Table 5 Comprehensive Breakdown of Performance Metrics on Faceforensics++

Method	Accuracy	Precision	Recall	F1-Score
Siamese-based	0.944	0.957	0.931	0.944
Meso4	0.928	0.942	0.914	0.928
Capsule	0.921	0.934	0.908	0.921
Our Proposed Method	0.933	0.945	0.921	0.933

The minimal performance gap between laboratory conditions and real-world scenarios further cements the method's practical relevance. Although there remains a marginal scope for false negatives, the current framework stands as a good and comprehensive solution to the deepfake menace.

5 Discussion

The domain of deepfake detection is rapidly evolving, underpinned by the alarming sophistication of generative methods. In this light, our findings offer a novel perspective on the integration of audio and visual data for more accurate detection.

1. **Strengths of the Audio-Visual Fusion**

Our proposed audio-visual fusion framework demonstrates several key strengths. First and foremost, by holistically analyzing both auditory and visual discrepancies in videos, the model is better equipped to detect incongruities often overlooked when either modality is considered in isolation. The hybrid fusion strategy consolidates the advantages of early and late fusion, optimizing feature integration and ensuring the final decision-making layer has a richer information pool.

2. **Limitations**

Table 6 Real-world Evaluation Results

Scenario	Accuracy	Notes
Low-light Condition	0.926	Model faced challenges due to shadows and poor quality
Noisy Background Audio	0.947	Model excelled in isolating voice from background noise

In the course of our study, we identified some limitations that can be explored further. We used a composite deepfake dataset, which may not cover all deepfake scenarios, indicating a need for more diverse datasets. Our complex deep-learning framework, while effective, requires substantial computational resources, suggesting a need for optimization. Lastly, our framework was tested under specific conditions, but a comprehensive performance analysis against novel deepfake methods was not conducted, highlighting the need for robust testing. These findings provide directions for future deepfake detection research.

6 Conclusion

Deepfakes, with their potential to distort truth and mislead the public, pose a significant threat in this digital age. Addressing this challenge, our proposed audio-visual fusion methodology offers a powerful countermeasure to detect and combat these manipulations. By concurrently considering both visual and auditory cues, this study reaffirms the significance of a multi-modal approach in understanding and recognizing deepfakes with heightened accuracy.

The framework's robust performance, as evidenced by our evaluations, suggests that the intricacies of authentic human behavior — the synchronization of voice, lip movements, facial expressions, and subtle auditory nuances — are challenging for even the most sophisticated deepfake algorithms to replicate flawlessly. These incongruities, no matter how minute, become the Achilles' heel of deepfakes when subjected to our rigorous detection framework.

Moreover, the potential applications of this framework are vast. From social media platforms to news agencies, the necessity to validate the authenticity of digital content is paramount. Implementing such an advanced detection system can act as a bulwark against misinformation campaigns, ensuring the veracity of digital interactions.

In conclusion, while the arms race between deepfake generation and detection will likely continue to escalate, our research demonstrates that innovation, especially through multi-modal fusion strategies, can keep us one step ahead. As technology advances and the line between reality and fabrication blurs further, efforts like ours become critical in safeguarding truth in the digital realm.

7 Future Research Directions

The rapidly evolving landscape of deepfake generation underscores the constant need for adaptive and forward-thinking detection techniques. While our multi-modal fusion

strategy has proven effective, several avenues remain unexplored and hold promising potential.

1. **Higher-Dimensional Data**: Our framework harnesses audio and visual cues. Incorporating additional data dimensions, such as physiological signals (e.g., heart rate or skin temperature variations) captured through advanced sensors, might yield even more refined detection accuracy.
2. **Temporal Analysis**: Deepfake algorithms might stumble in maintaining consistent realism throughout a video's duration. Investigating inconsistencies over time rather than isolated frames might offer another layer of discernment.
3. **Cross-Modal Interactions**: While we've fused audio and visual cues, a deeper understanding of their intricate interplay, particularly how one modality might affect the perception of another, could lead to more nuanced detection strategies.
4. **Ethical Implications**: Future research should also delve into the societal and ethical considerations of deepfake detection. As technologies advance, ensuring that they respect user privacy and don't inadvertently suppress legitimate content becomes paramount.

In sum, while our work provides a formidable stand against current deepfake challenges, it also sets the stage for a plethora of research opportunities. By anticipating and adapting to the next wave of synthetic media challenges, we can continue to champion the cause of digital truth.

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